

Stony Brook Institute at
Anhui University (SBIAHU)

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Applied Calculus IV

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Week 2

- 1.1 & 1.2 (review)
- 1.3 (models)
- 2.1

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Exercises 1.1 & 1.2

Linearity

In Problems 1–8 state the order of the given ordinary differential equation. Determine whether the equation is linear or nonlinear by matching it with (6).

1. $(1 - x)y'' - 4xy' + 5y = \cos x$

2. $x \frac{d^3y}{dx^3} - \left(\frac{dy}{dx}\right)^4 + y = 0$

3. $t^5y^{(4)} - t^3y'' + 6y = 0$

4. $\frac{d^2u}{dr^2} + \frac{du}{dr} + u = \cos(r + u)$

5. $\frac{d^2y}{dx^2} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$

6. $\frac{d^2R}{dt^2} = -\frac{k}{R^2}$

General solution

In Problems 21–24 verify that the indicated family of functions is a solution of the given differential equation. Assume an appropriate interval I of definition for each solution.

$$21. \frac{dP}{dt} = P(1 - P); \quad P = \frac{c_1 e^t}{1 + c_1 e^t}$$

$$22. \frac{dy}{dx} + 4xy = 8x^3; \quad y = 2x^2 - 1 + c_1 e^{-2x^2}$$

$$23. \frac{d^2 y}{dx^2} - 4 \frac{dy}{dx} + 4y = 0; \quad y = c_1 e^{2x} + c_2 x e^{2x}$$

$$24. x^3 \frac{d^3 y}{dx^3} + 2x^2 \frac{d^2 y}{dx^2} - x \frac{dy}{dx} + y = 12x^2;$$

$$y = c_1 x^{-1} + c_2 x + c_3 x \ln x + 4x^2$$

1.

$$\begin{aligned}\frac{dP}{dt} &= \frac{(1 + c_1 e^t) c_1 e^t - c_1 e^t \cdot c_1 e^t}{(1 + c_1 e^t)^2} = \frac{c_1 e^t}{1 + c_1 e^t} \frac{[(1 + c_1 e^t) - c_1 e^t]}{1 + c_1 e^t} \\ &= \frac{c_1 e^t}{1 + c_1 e^t} \left[1 - \frac{c_1 e^t}{1 + c_1 e^t} \right] = P(1 - P).\end{aligned}$$

2.

$$y = 2x^2 - 1 + c_1 e^{-2x^2} \implies \frac{dy}{dx} = 4x - 4x c_1 e^{-2x^2} \implies$$

$$\frac{dy}{dx} + 4xy = 4x - 4x c_1 e^{-2x^2} + 8x^3 - 4x + 4c_1 x e^{-x^2} = 8x^3$$

Initial Value Problem

In Problems 1 and 2, $y = 1/(1 + c_1 e^{-x})$ is a one-parameter family of solutions of the first-order DE $y' = y - y^2$. Find a solution of the first-order IVP consisting of this differential equation and the given initial condition.

1. $y(0) = -\frac{1}{3}$

2. $y(-1) = 2$

1. Solving $-\frac{1}{3} = \frac{1}{1 + c_1} \implies c_1 = -4 \implies y = \frac{1}{1 - 4e^{-x}}$

2. Solving $2 = \frac{1}{1 + c_1 e} \implies c_1 = -\frac{1}{2}e^{-1} \implies y = \frac{2}{2 - e^{-(x+1)}}$

Initial Value Problem

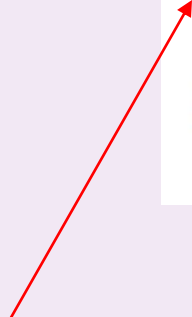
In Problems 3–6, $y = 1/(x^2 + c)$ is a one-parameter family of solutions of the first-order DE $y' + 2xy^2 = 0$. Find a solution of the first-order IVP consisting of this differential equation and the given initial condition. Give the largest interval I over which the solution is defined.

3. $y(2) = \frac{1}{3}$

4. $y(-2) = \frac{1}{2}$

5. $y(0) = 1$

6. $y\left(\frac{1}{2}\right) = -4$



Letting $x = 2$ and solving $\frac{1}{3} = \frac{1}{4 + c} \implies c = -1 \implies y = \frac{1}{x^2 - 1}$.

This solution is defined on the interval $(1, \infty)$.

Initial Value Problem

In Problems 11–14, $y = c_1e^x + c_2e^{-x}$ is a two-parameter family of solutions of the second-order DE $y'' - y = 0$. Find a solution of the second-order IVP consisting of this differential equation and the given initial conditions.

11. $y(0) = 1, \quad y'(0) = 2$

From the initial conditions, we have the system of equations:

$$\begin{cases} c_1 + c_2 = 1 \\ c_1 - c_2 = 2 \end{cases}$$

$$c_1 = \frac{3}{2}, \quad c_2 = -\frac{1}{2}$$

The solution of the initial - value problem is $y = \frac{3}{2}e^x - \frac{1}{2}e^{-x}$.

Existence & Uniqueness

In Problems 25–28 determine whether Theorem 1.2.1 guarantees that the differential equation $y' = \sqrt{y^2 - 9}$ possesses a unique solution through the given point.

25. $(1, 4)$

26. $(5, 3)$

27. $(2, -3)$

28. $(-1, 1)$

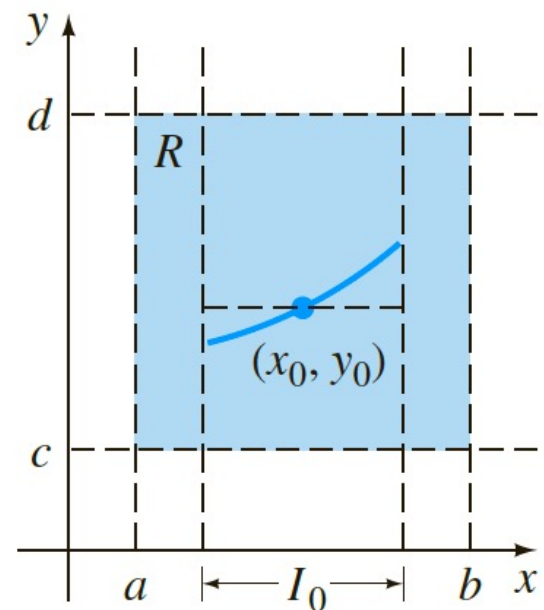
THEOREM 1.2.1 Existence of a Unique Solution

Let R be a rectangular region in the xy -plane defined by $a \leq x \leq b$, $c \leq y \leq d$ that contains the point (x_0, y_0) in its interior. If $f(x, y)$ and $\partial f / \partial y$ are continuous on R , then there exists some interval $I_0: (x_0 - h, x_0 + h)$, $h > 0$, contained in $[a, b]$, and a unique function $y(x)$, defined on I_0 , that is a solution of the initial-value problem (2).

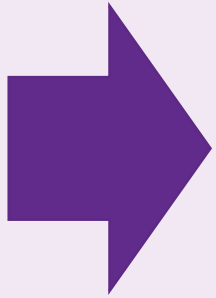


$$\frac{dy}{dx} = f(x, y)$$

$$y(x_0) = y_0$$



To guarantees that $y' = f(x, y)$ possesses a unique solution through a given point



- ✓ The point must be in the domain of $f(x, y)$ and $\frac{\partial f(x, y)}{\partial y}$
- ✓ The point must have neighborhood

Existence & Uniqueness

In Problems 25–28 determine whether Theorem 1.2.1 guarantees that the differential equation $y' = \sqrt{y^2 - 9}$ possesses a unique solution through the given point.

25. $(1, 4)$

26. $(5, 3)$

27. $(2, -3)$

28. $(-1, 1)$

The domain of the $f(x, y)$ is $(-\infty, -3] \cup [3, \infty)$

but we need y in $(-\infty, -3) \cup (3, \infty)$

- ✓ The point $(1,4)$ satisfies $y > 3$, so the DE has a unique solution passing through $(1,4)$.
- ✓ The point $(5,3)$ does not satisfy $y < -3$ or $y > 3$, so there's no guarantee of a unique solution through $(5,3)$.
- ✓ The point $(2,-3)$ does not satisfy $y < -3$ or $y > 3$, so there's no guarantee of a unique solution through $(2,-3)$.
- ✓ The point $(-1,1)$ does not satisfy $y < -3$ or $y > 3$, so there's no guarantee of a unique solution through $(-1,1)$.

Existence & Uniqueness

In Problems 17–24 determine a region of the xy -plane for which the given differential equation would have a unique solution whose graph passes through a point (x_0, y_0) in the region.

17. $\frac{dy}{dx} = y^{2/3}$

18. $\frac{dy}{dx} = \sqrt{xy}$

19. $x \frac{dy}{dx} = y$

20. $\frac{dy}{dx} - y = x$

21. $(4 - y^2)y' = x^2$

22. $(1 + y^3)y' = x^2$

23. $(x^2 + y^2)y' = y^2$

24. $(y - x)y' = y + x$

Solution

17. $f(x, y) = y^{2/3} \implies \frac{\partial f}{\partial y} = \frac{2}{3}y^{-1/3}$. Unique solution where $y \neq 0$.

18. $f(x, y) = \sqrt{xy} \implies \frac{\partial f}{\partial y} = \frac{1}{2}\sqrt{\frac{x}{y}}$. Unique solution where $x > 0, y > 0$ or $x < 0, y < 0$.

19. $f(x, y) = \frac{y}{x} \implies \frac{\partial f}{\partial y} = \frac{1}{x}$. Unique solution where $x \neq 0$.

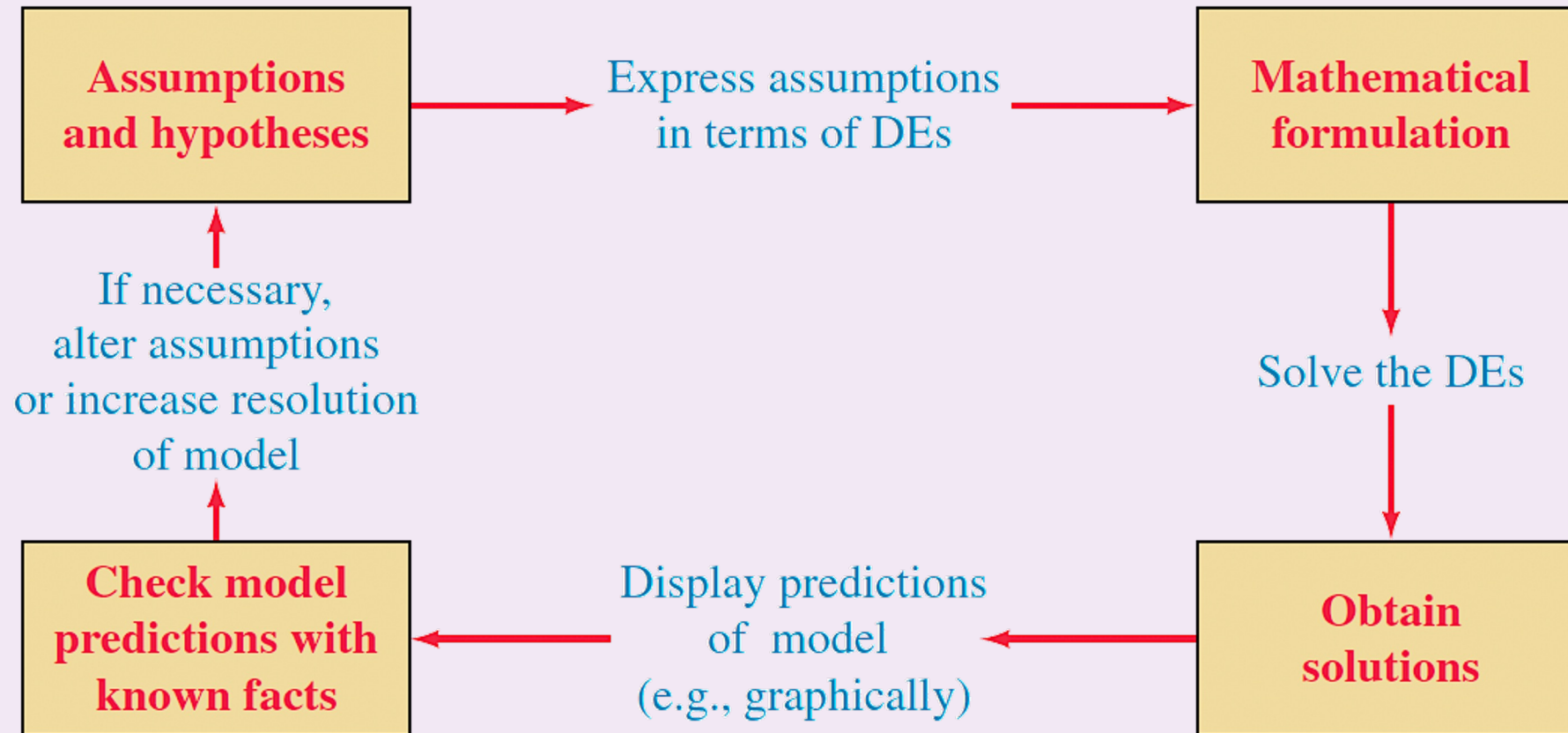
20. $f(x, y) = x + y \implies \frac{\partial f}{\partial y} = 1$. Unique solution in entire plane.

Differential Equations as Mathematical Models

Mathematical models represent real-world systems to achieve specific goals.

Model construction has two core steps:

- identifying key **variables** that affect the system, determining its resolution
- making reasonable assumptions and including relevant empirical laws (formulation)



Example (population)

Thomas Malthus (18th century)

He made an early math
model for population growth

He understood that a population's
growth rate at a time is proportional
to its total size

$$\frac{dP}{dt} \propto P$$

t is time population



$$\frac{dP}{dt} = kP$$

k is a constant

- ✓ It's a simple version (with no factors which may influence the population)
- ✓ This simple model practically worked for US population 1790–1860
- ✓ It can be used for small populations over short times, like bacteria in a dish.

Example (in finance)

This ODE describes the growth of capital S when an annual rate of interest r is compounded continuously.

$$\frac{dS}{dt} = rS$$



solution

$$S(t) = S_0 e^{rt}$$



Initial value, $S(0) = S_0$



$$S(1) = 1000e^{0.05 \times 1} \approx 1000 \times 1.0513 = 1051.3 \text{ dollars.}$$

$$S(2) = 1000e^{0.05 \times 2} \approx 1000 \times 1.1052 = 1105.2 \text{ dollars.}$$

Example (spread of a disease)

The disease spread rate is assumed proportional to interactions between infected and unexposed people

We are looking for a model to describe this dynamic and help us to estimate later number of infected people

Step one - Define variables

The disease spread rate is assumed proportional to interactions between infected and unexposed

people

$$\frac{dx}{dt}$$



$$x(t)$$


$$y(t)$$

Step two – Define the formula

$$\frac{dx}{dt} = kxy,$$


k is a constant

For a community with a fixed population n and one initial infected person, $x + y = n + 1$.



$$y = n + 1 - x$$



$$\frac{dx}{dt} = kx(n + 1 - x).$$

with the initial condition $x(0) = 1$.

From now on, we focus on

Solving Differential Equations

Methods & Techniques

First-Order Differential Equations

- 
- 2.1** Solution Curves Without a Solution
 - 2.2** Separable Equations
 - 2.3** Linear Equations
 - 2.4** Exact Equations
 - 2.5** Solutions by Substitutions
 - 2.6** A Numerical Method

2.1 Solution Curves Without a Solution



We want to analyze the system without having the solution

In a first-order differential equation,

$$\frac{dy}{dx} = f(x, y)$$

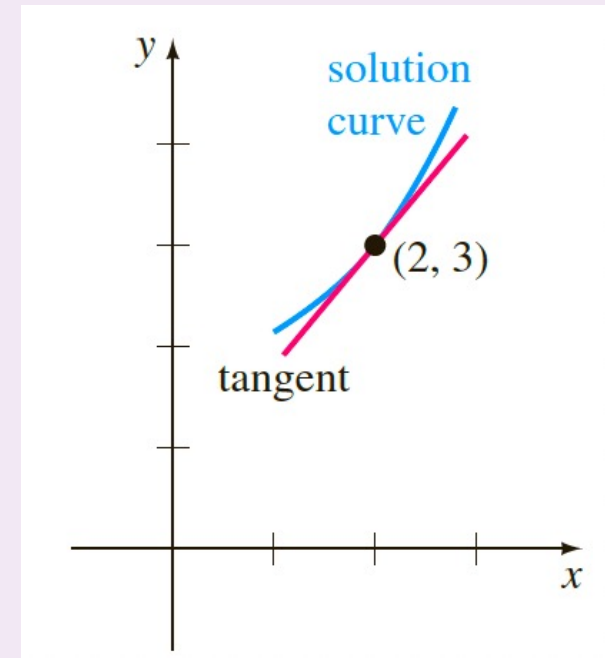
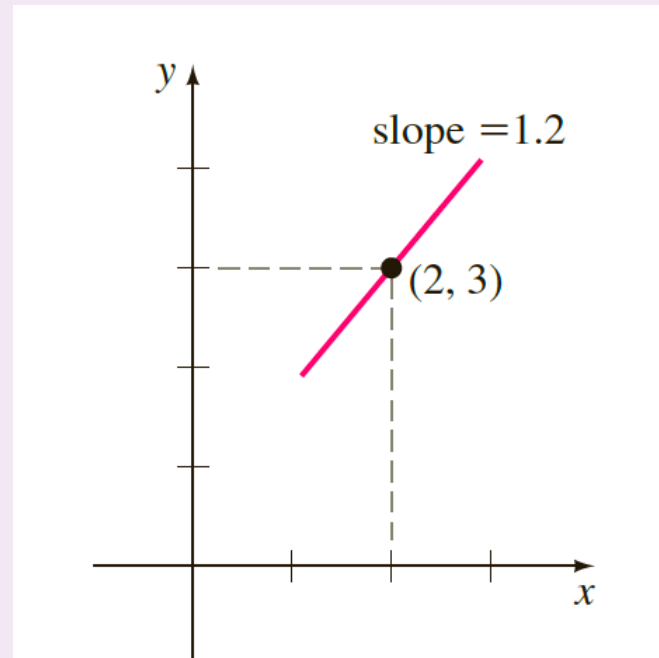


This value shows **the slope** of the solution at time t

Example:

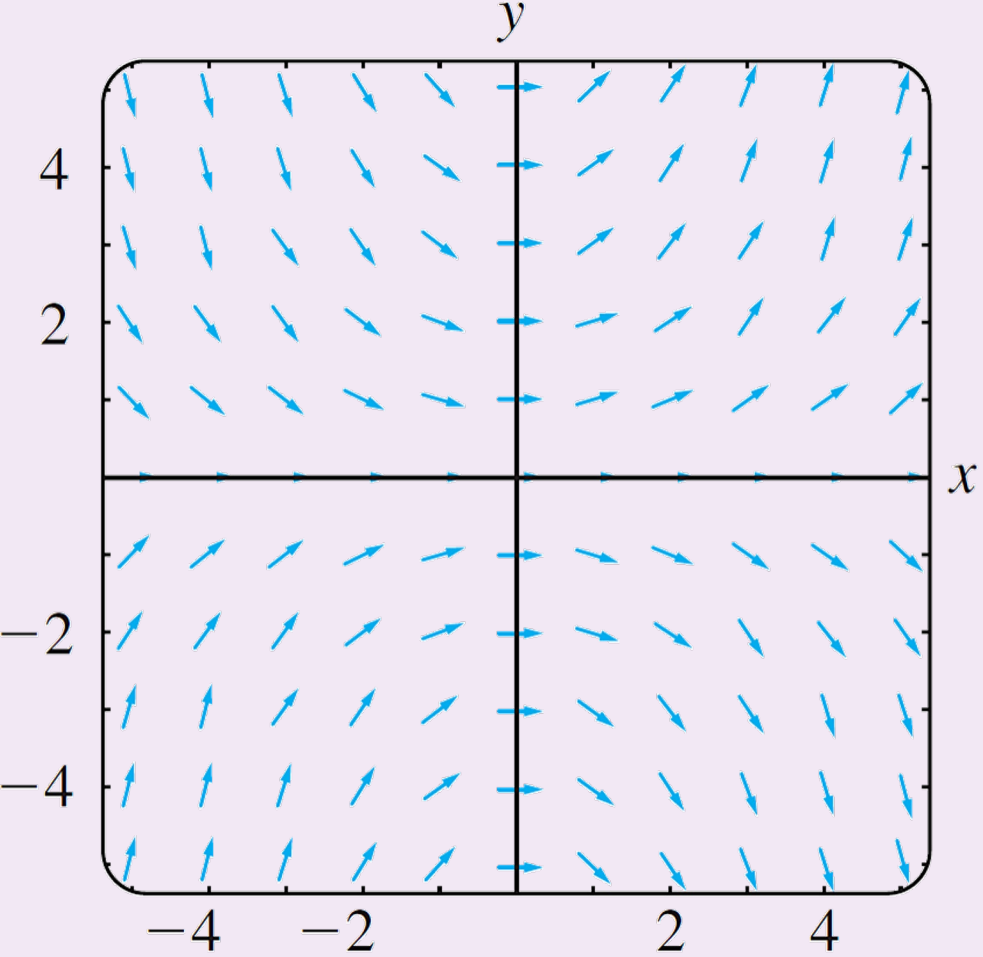
$$\frac{dy}{dx} = 0.2xy$$

$$f(x, y) = f(2, 3) = 1.2$$



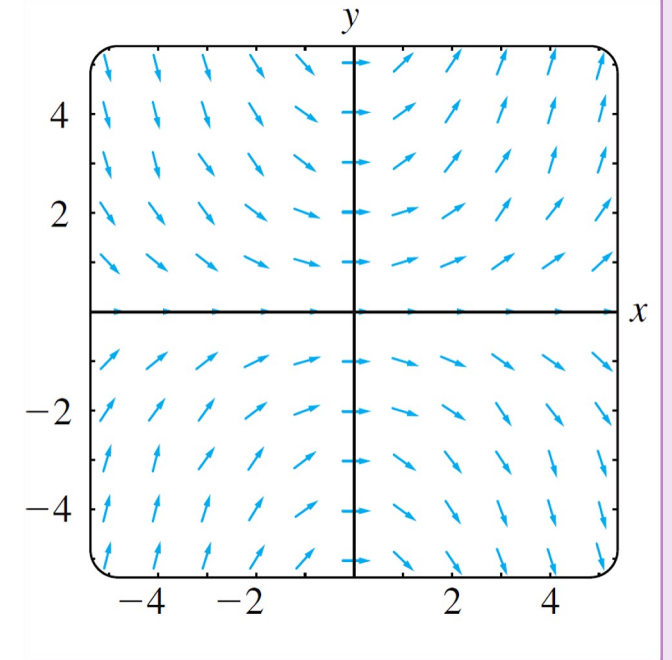
Continue it for more points:

Region	x	y	xy	$\frac{dy}{dx} = 0.2xy$
$x > 0, y > 0$	Positive	Positive	Positive	Positive
$x > 0, y < 0$	Positive	Negative	Negative	Negative
$x < 0, y > 0$	Negative	Positive	Negative	Negative
$x < 0, y < 0$	Negative	Negative	Positive	Positive
$x = 0$ or $y = 0$	—	—	0	0



❖ It is called **direction fields** of differential equation $\frac{dy}{dx} = 0.2xy$

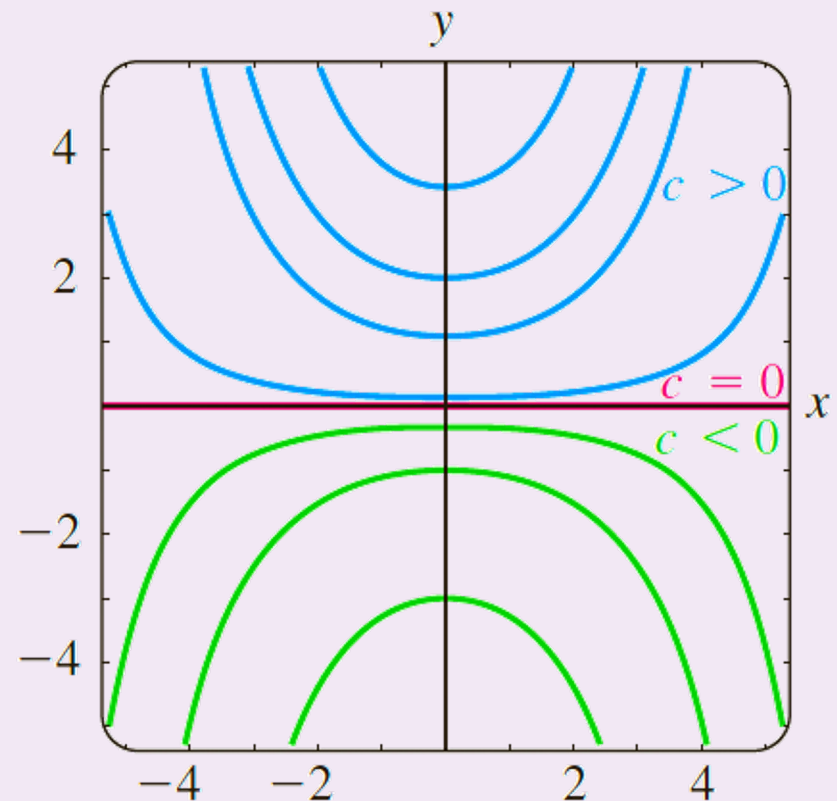
❖ It visually shows the shape of solution curves, and the key behaviors at a glance



Example: Verify that $y = ce^{0.1x}$ is general solution of

$$\frac{dy}{dx} = 0.2xy$$

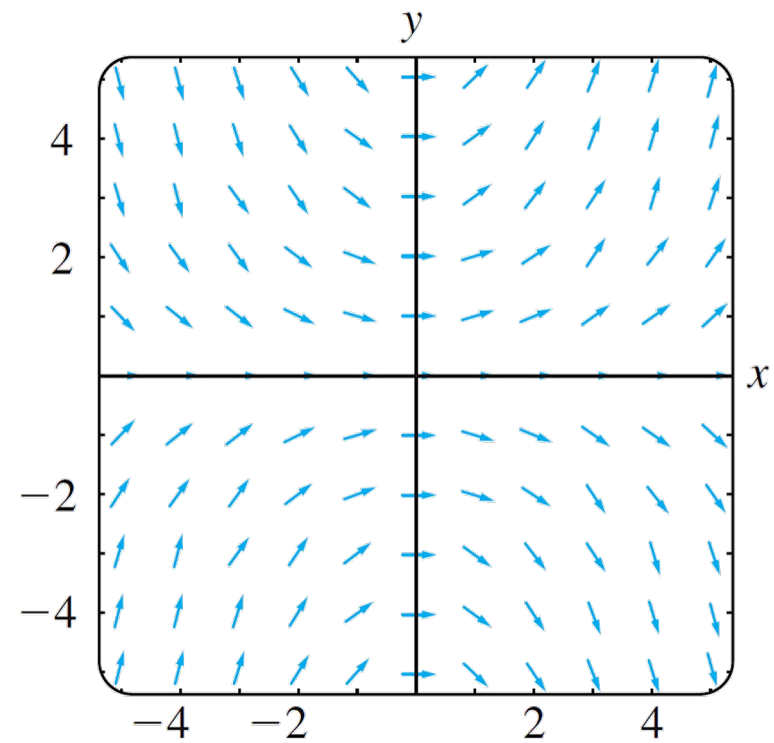
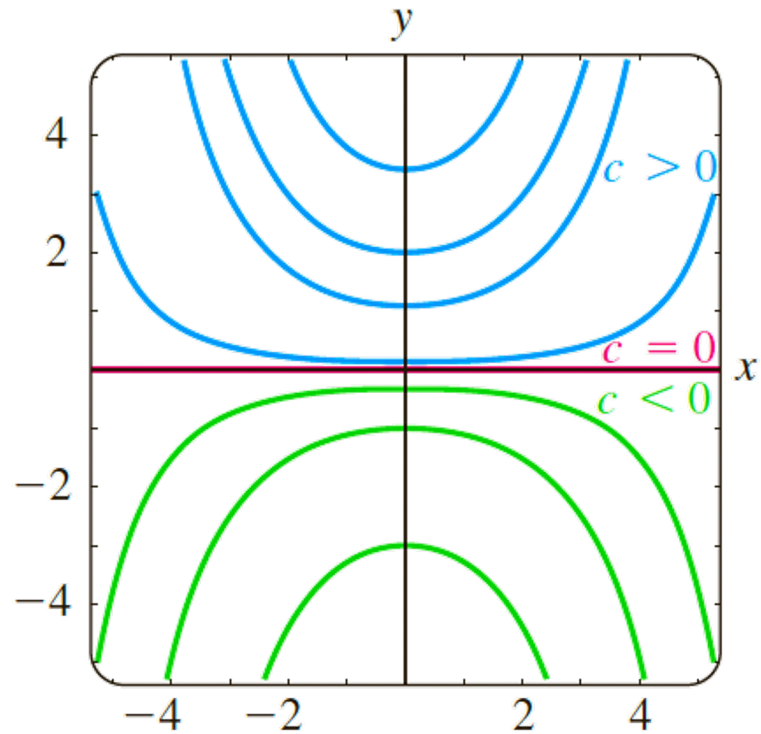
on the interval $(-\infty, \infty)$.



Solution Curves

vs

Direction Field



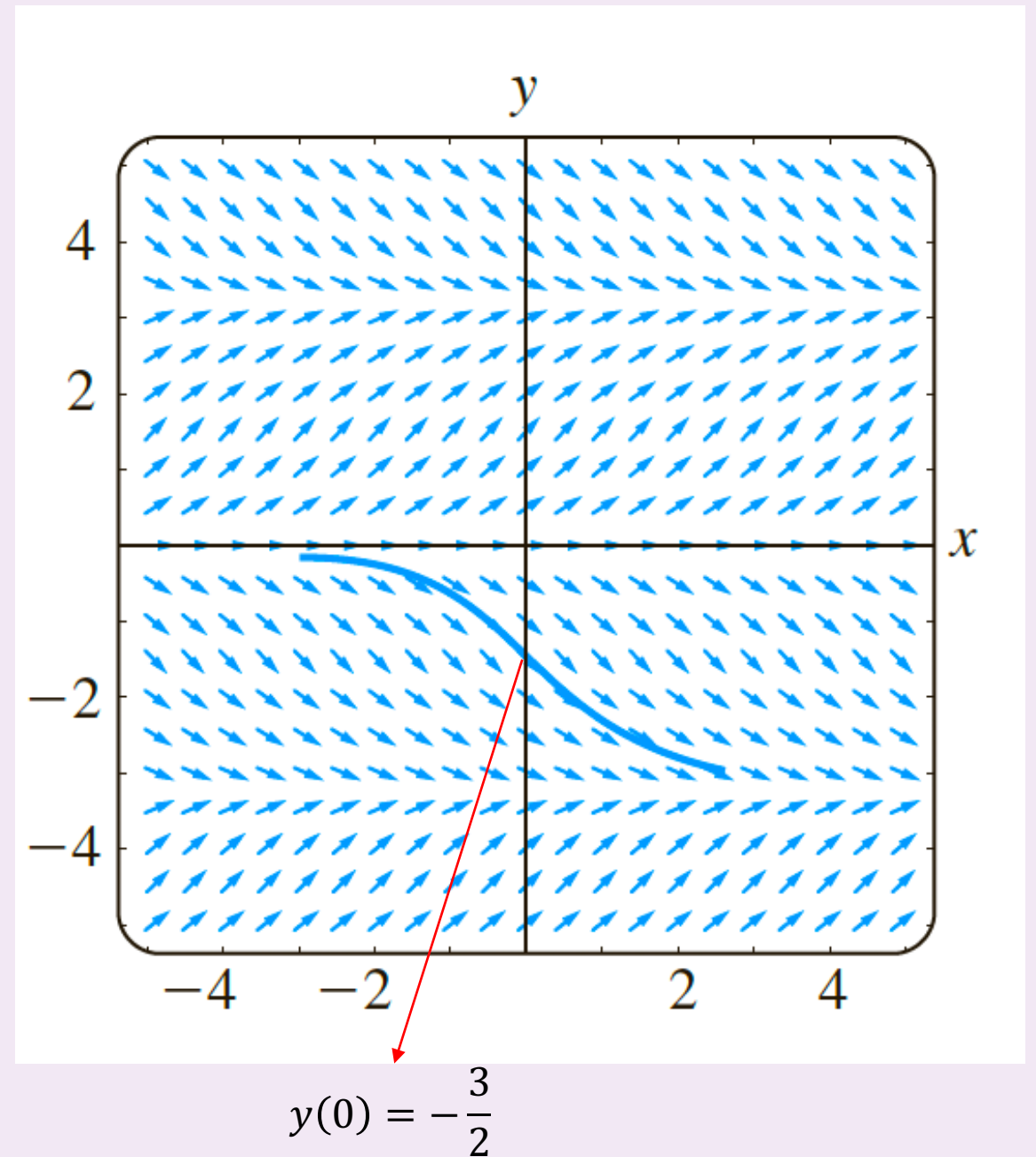
$$\frac{dy}{dx} = 0.2xy \rightarrow y = ce^{0.1x} \text{ (general solution)}$$

Example: Use direction field to sketch an approximate solution curve
for the initial-value problem

$$\frac{dy}{dx} = \sin y$$

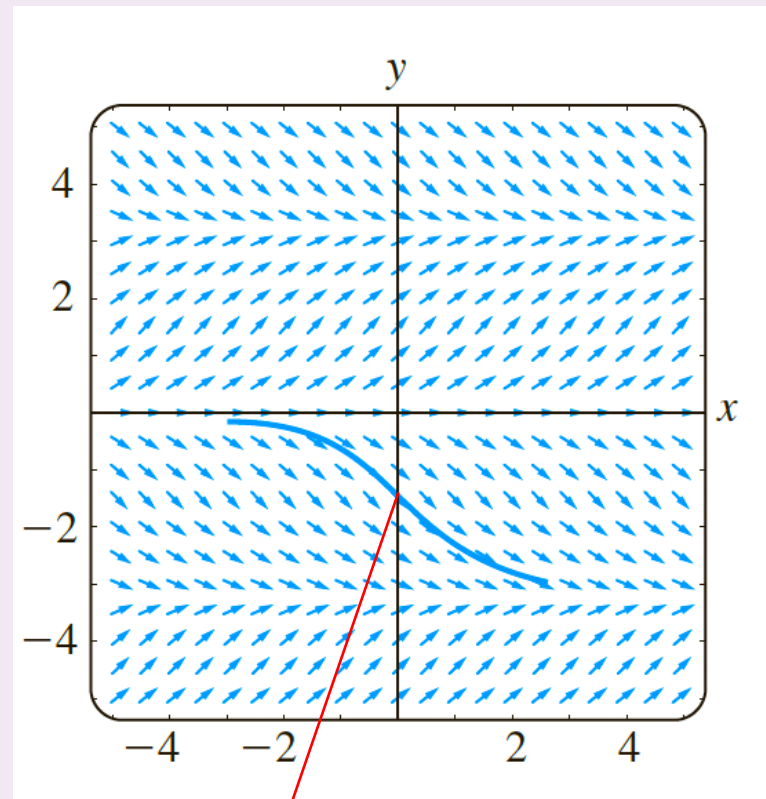
$$y(0) = -\frac{3}{2}$$

Interval (from $y = -3\pi$ to $y = 3\pi$)	Sign of $\sin y$
$-3\pi < y < -2\pi$	-
$-2\pi < y < -\pi$	+
$-\pi < y < 0$	-
$0 < y < \pi$	+
$\pi < y < 2\pi$	-
$2\pi < y < 3\pi$	+



and more details

Sign of $\sin y$	Interval range (n is an integer)	Example Values of y
0	$y = n\pi$ (Special Lines)	$y = 0, y = \pi, y = -\pi$
+	$2n\pi < y < (2n + 1)\pi$	$0 < y < \pi$ (when $n = 0$), $2\pi < y < 3\pi$ (when $n = 1$)
-	$(2n - 1)\pi < y < 2n\pi$	$-\pi < y < 0$ (when $n = 0$), $-2\pi < y < -\pi$ (when $n = -1$)



$$y(0) = -\frac{3}{2}$$

Autonomous First-Order DEs

$$\frac{dy}{dx} = f(y)$$

$$\frac{dy}{dx} = 1 + y^2$$

Nonautonomous First-Order DEs

$$\frac{dy}{dx} = f(x, y)$$

$$\frac{dy}{dx} = 0.2xy$$

Exercises 2.1

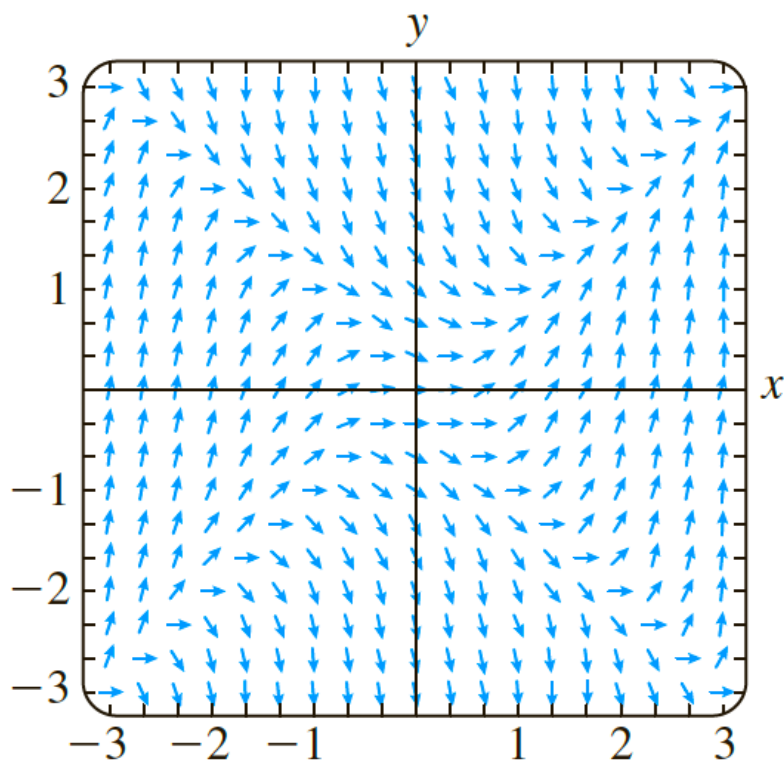
1. $\frac{dy}{dx} = x^2 - y^2$

(a) $y(-2) = 1$

(b) $y(3) = 0$

(c) $y(0) = 2$

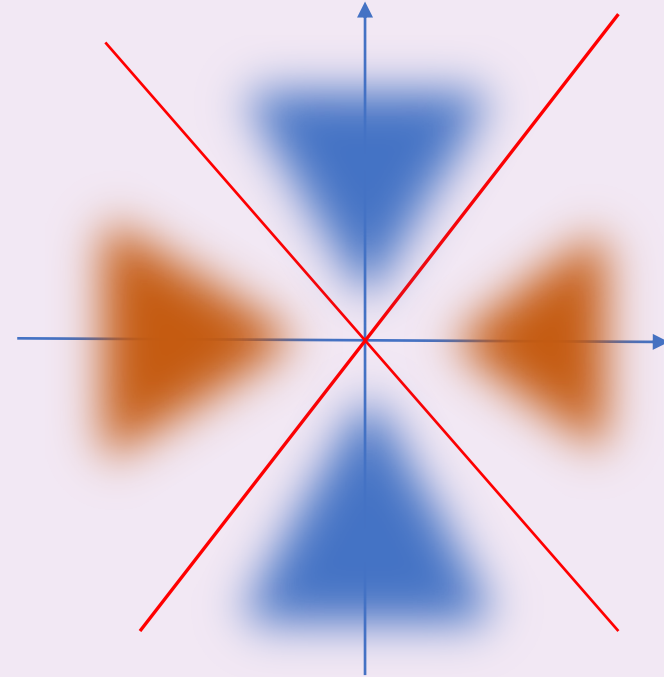
(d) $y(0) = 0$

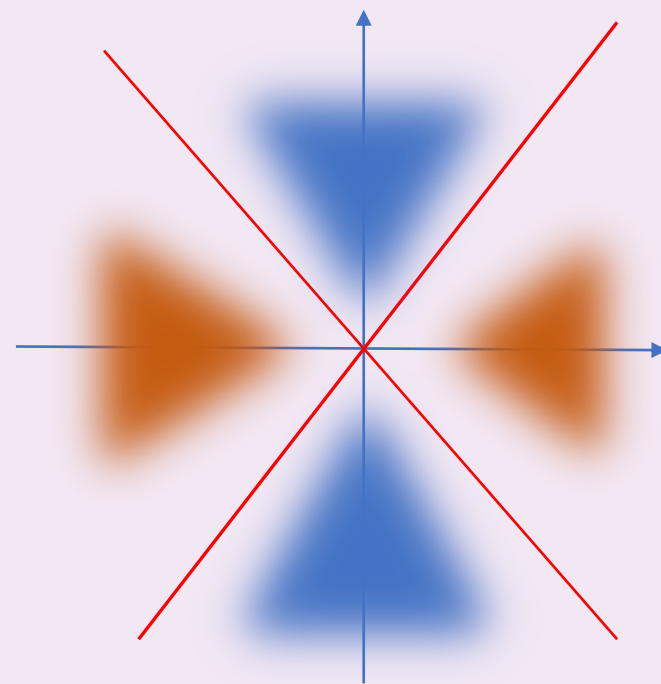
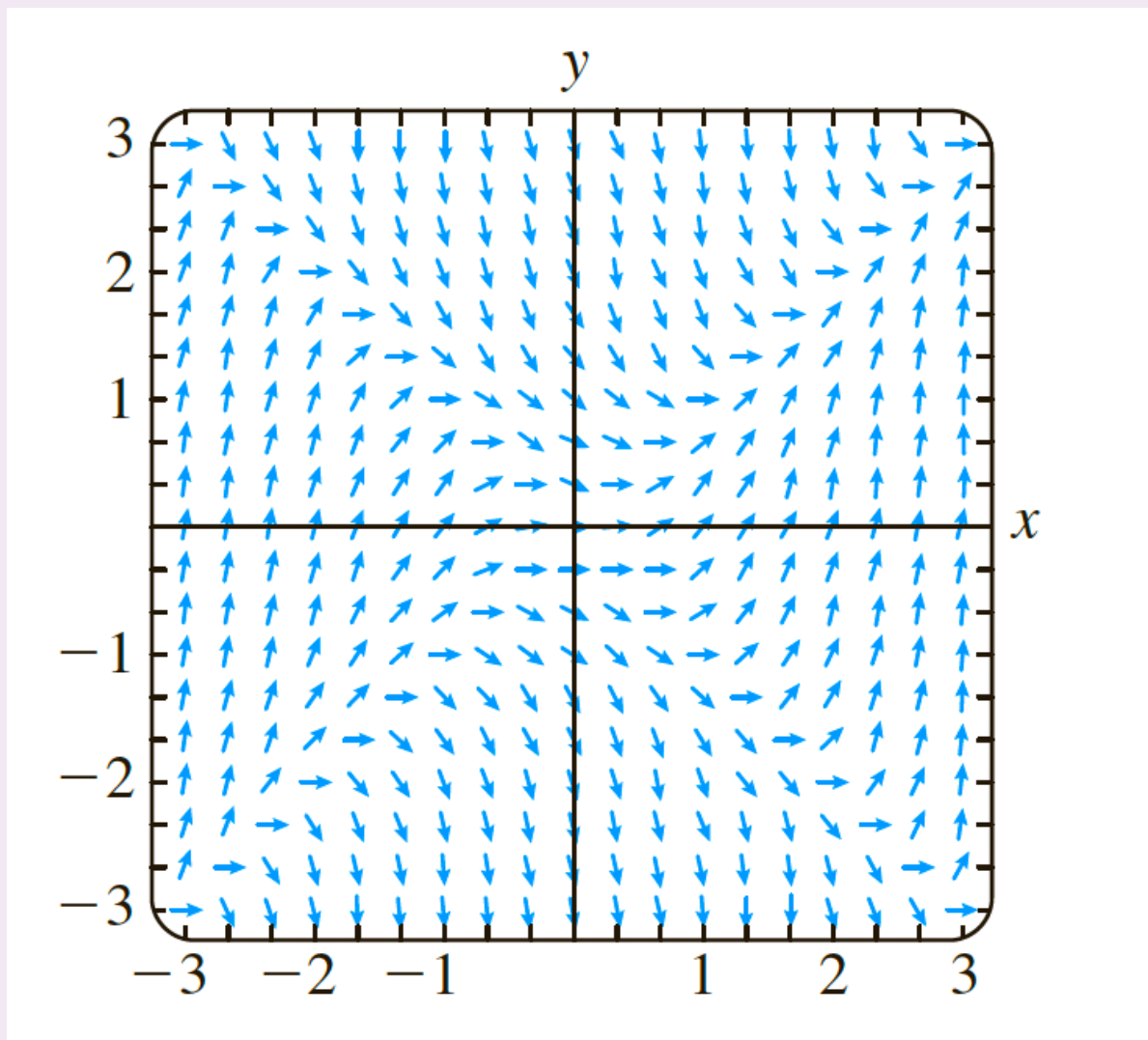


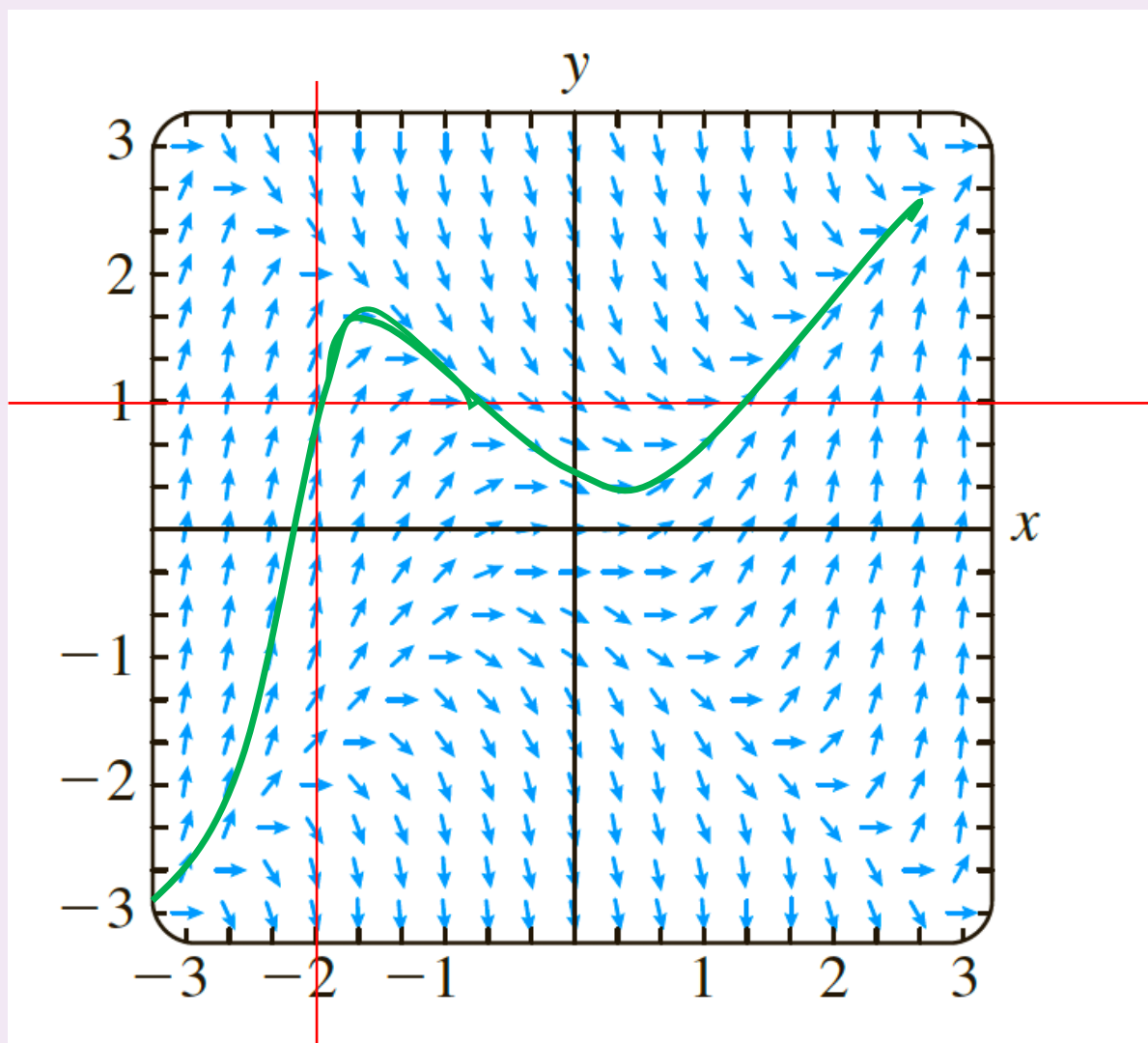
In Problems 1–4,

- Reproduce the given computer-generated direction field
- Sketch an approximate solution curve that passes through each of the indicated points.

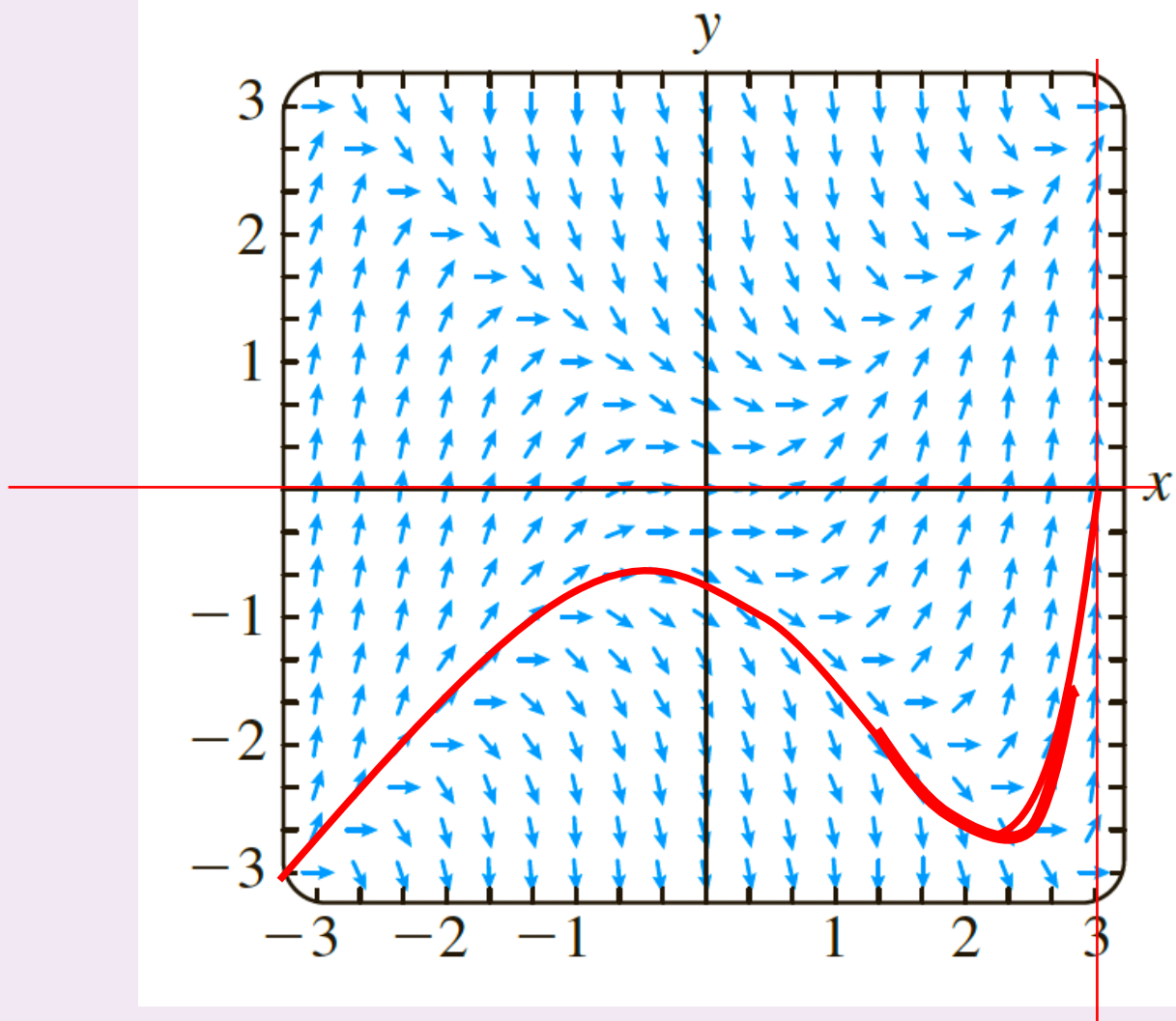
- **On $y = x$ or $y = -x$:** $x^2 - y^2 = 0$, so slope = 0 (horizontal segments).
- **Inside $|y| < |x|$:** $x^2 - y^2 > 0$, so slope > 0 (rising segments).
- **Outside $|y| > |x|$:** $x^2 - y^2 < 0$, so slope < 0 (falling segments).



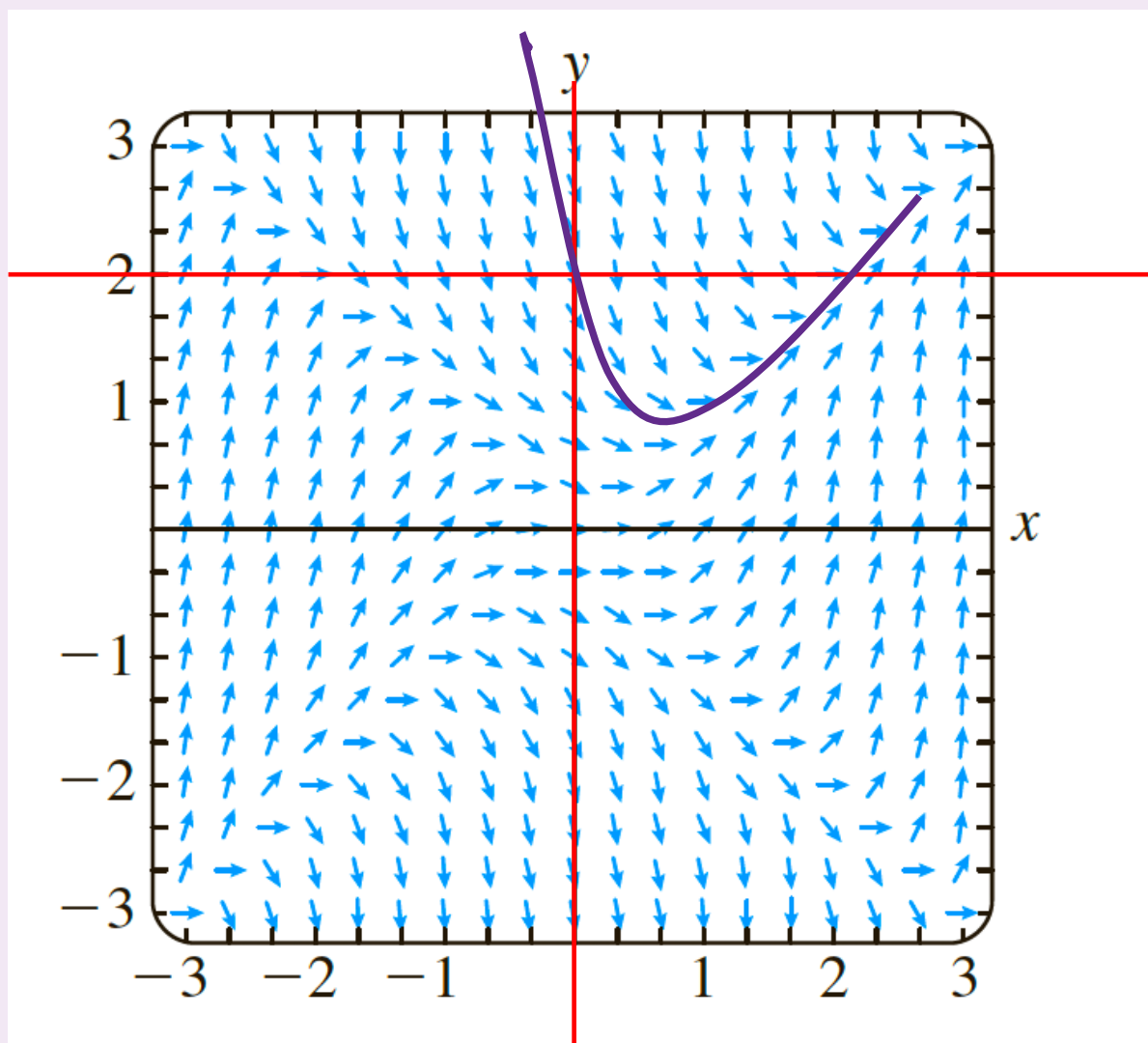




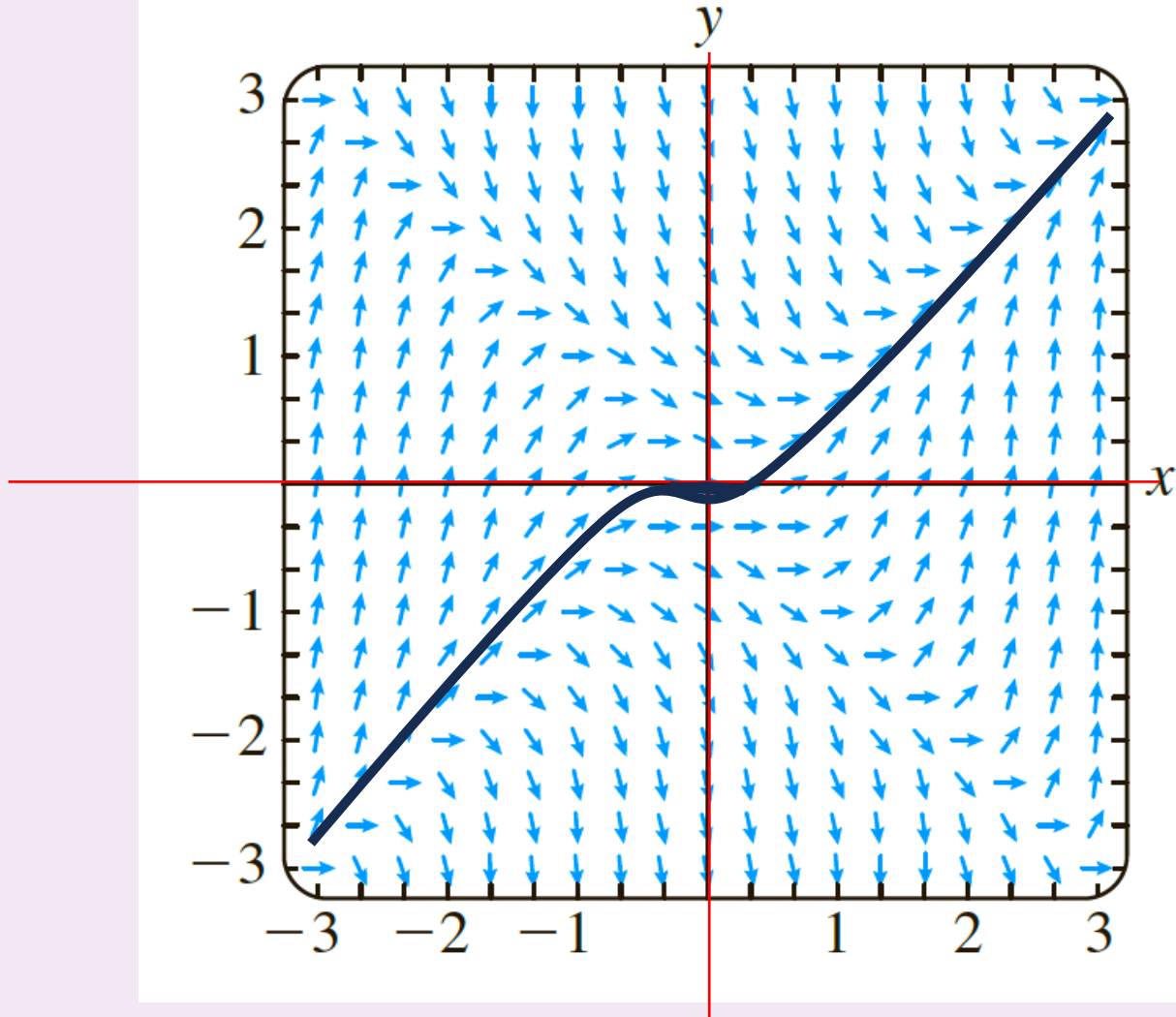
(a) $y(-2) = 1$



(b) $y(3) = 0$

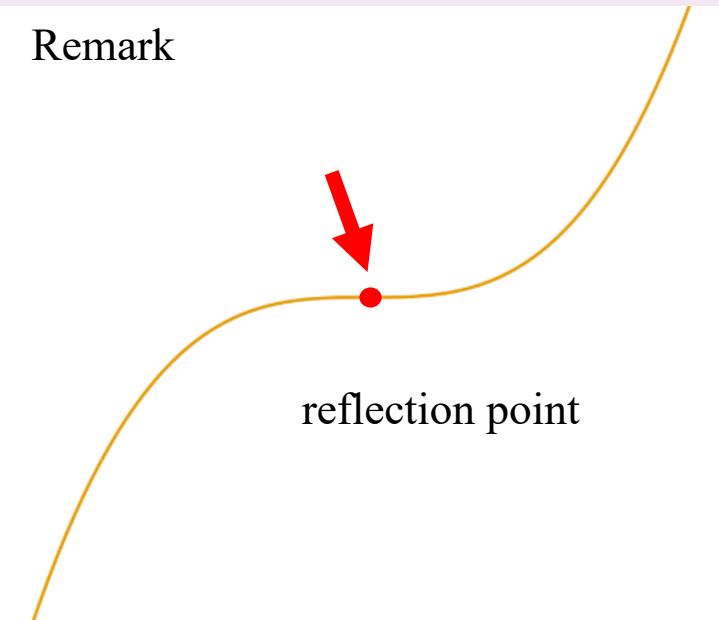


(c) $y(0) = 2$



(d) $y(0) = 0$

Remark



A diagram showing a smooth, upward-curving orange line. A red dot is placed on the curve at a point where the slope is nearly zero. A red arrow points down towards this dot. The text "reflection point" is written below the dot.

reflection point

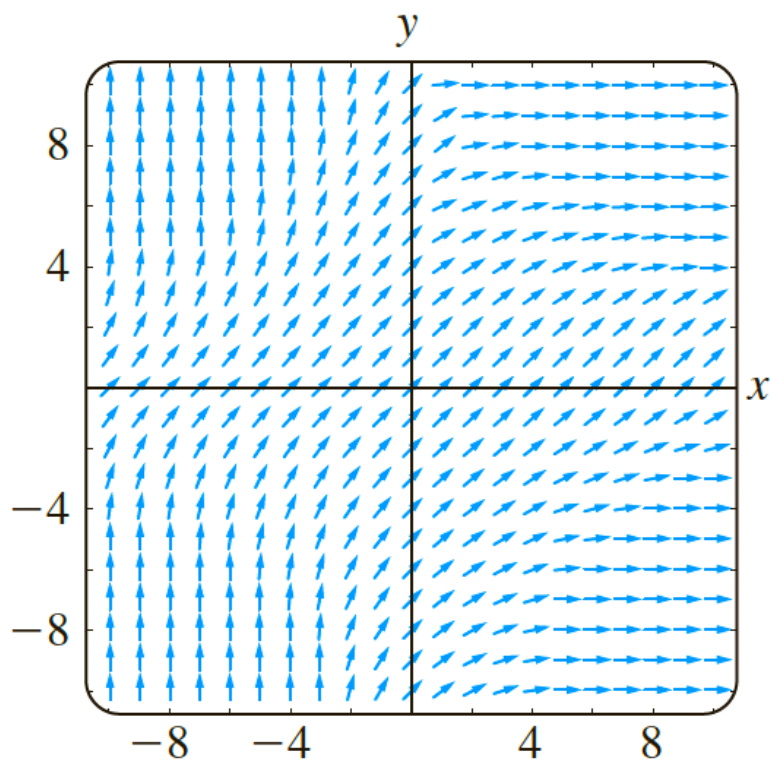
2. $\frac{dy}{dx} = e^{-0.01xy^2}$

(a) $y(-6) = 0$

(b) $y(0) = 1$

(c) $y(0) = -4$

(d) $y(8) = -4$

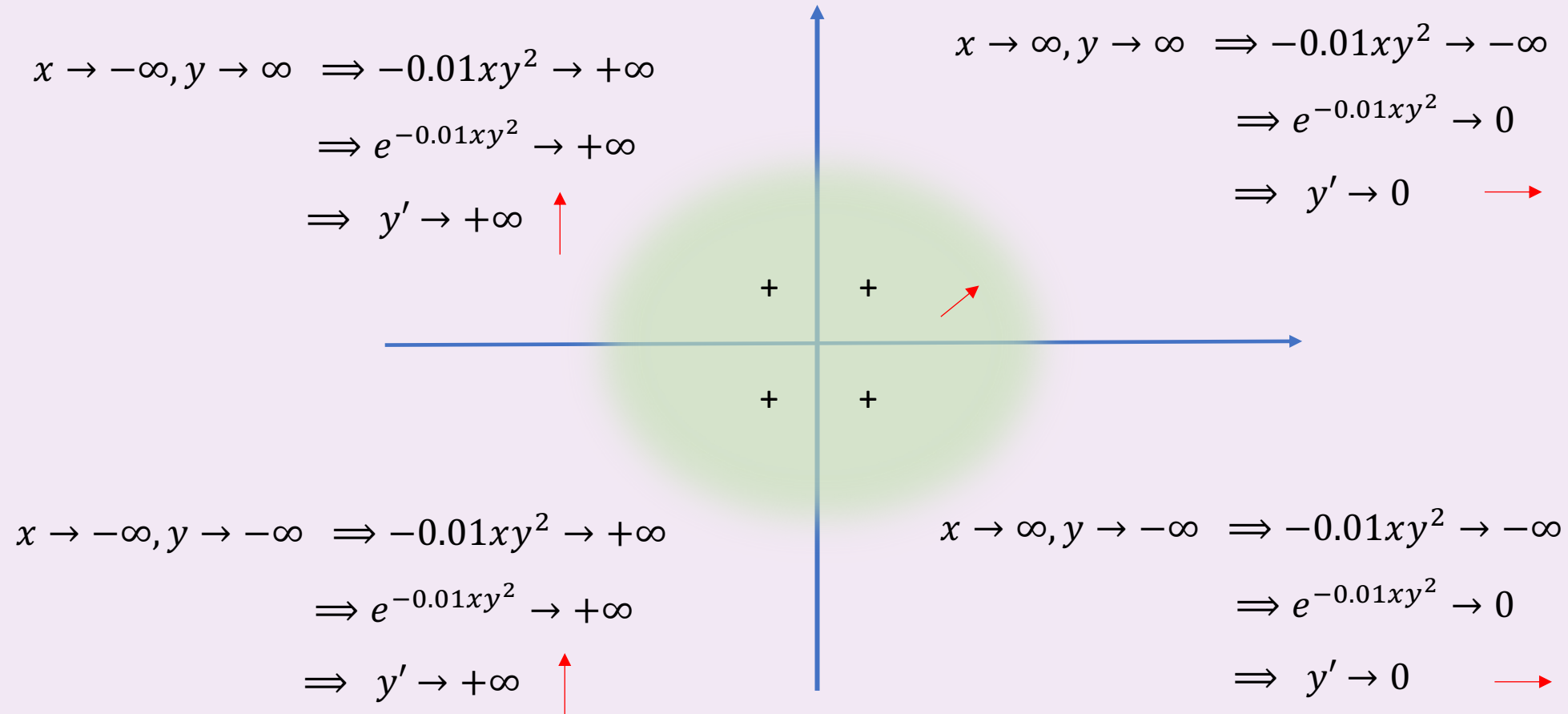


To draw the direction field, first we

should identify the sign of $e^{-0.01xy^2}$

(which equals y')

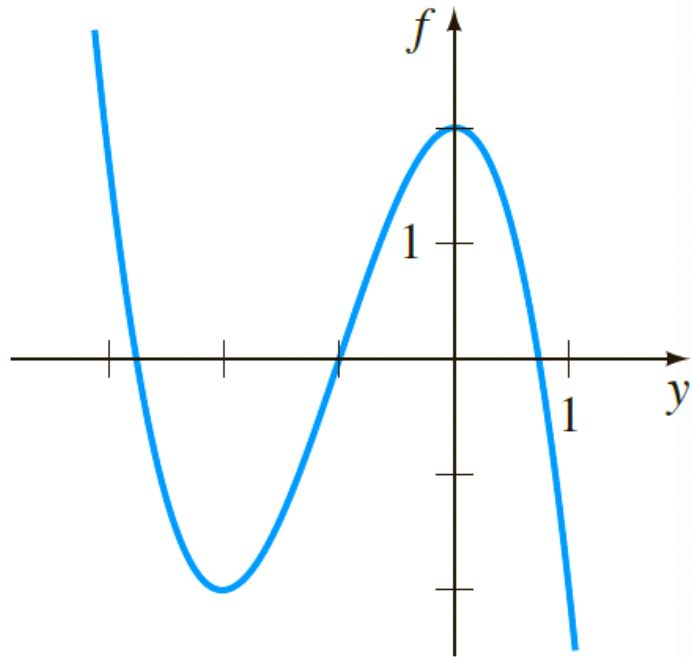
Identifying the sign of $e^{-0.01xy^2}$ (which equals y')



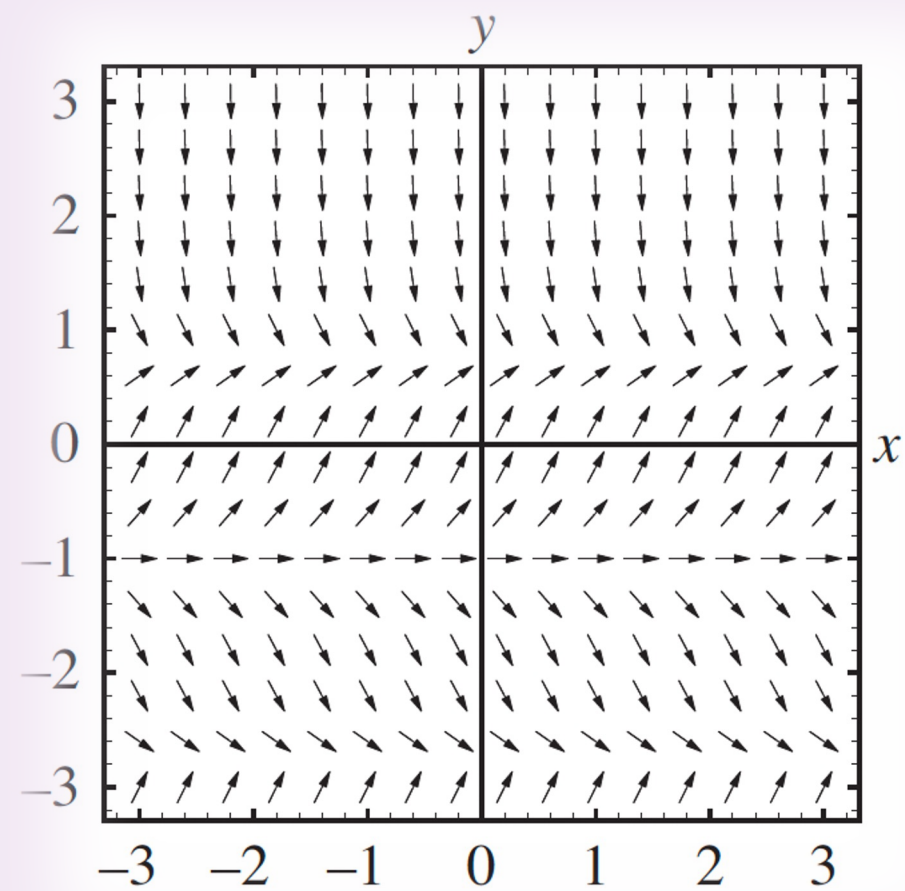
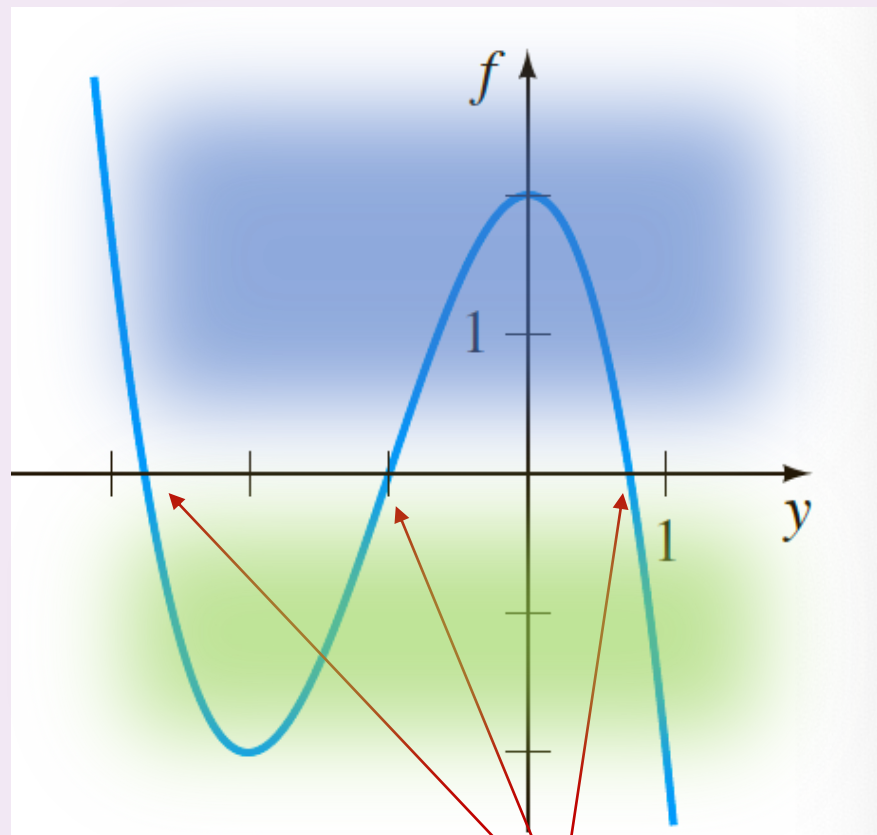
Challenge Yourself

By hand, sketch a direction field over an appropriate grid for $\frac{dy}{dx} = f(y)$

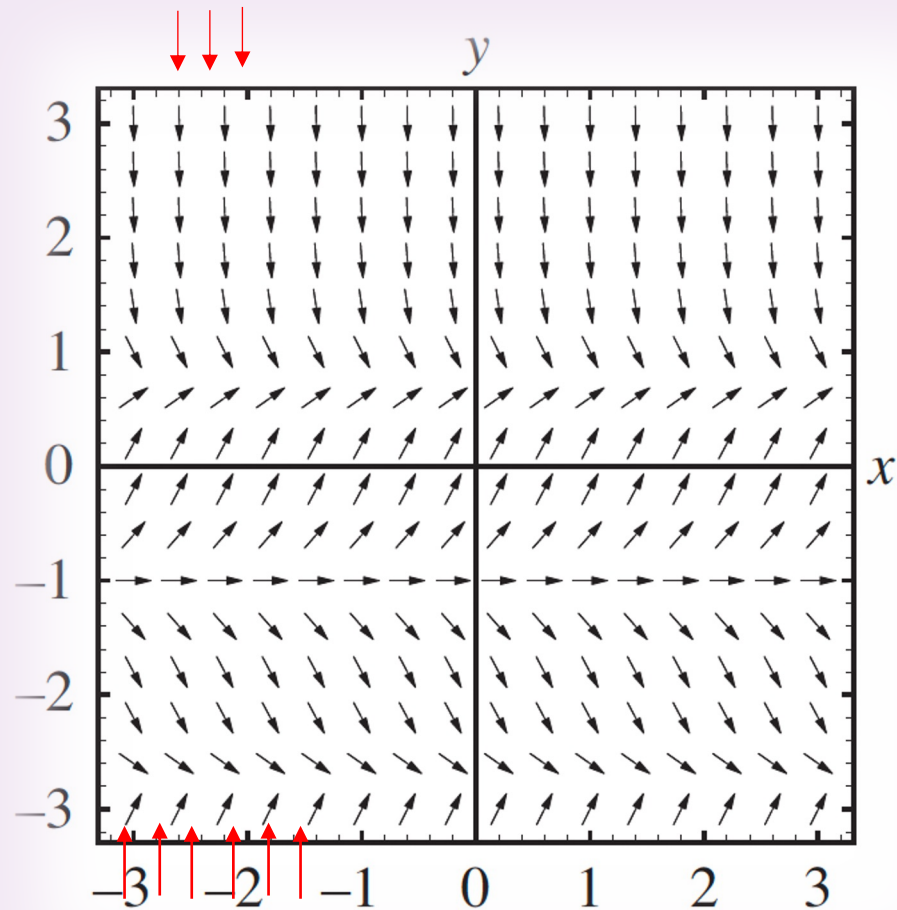
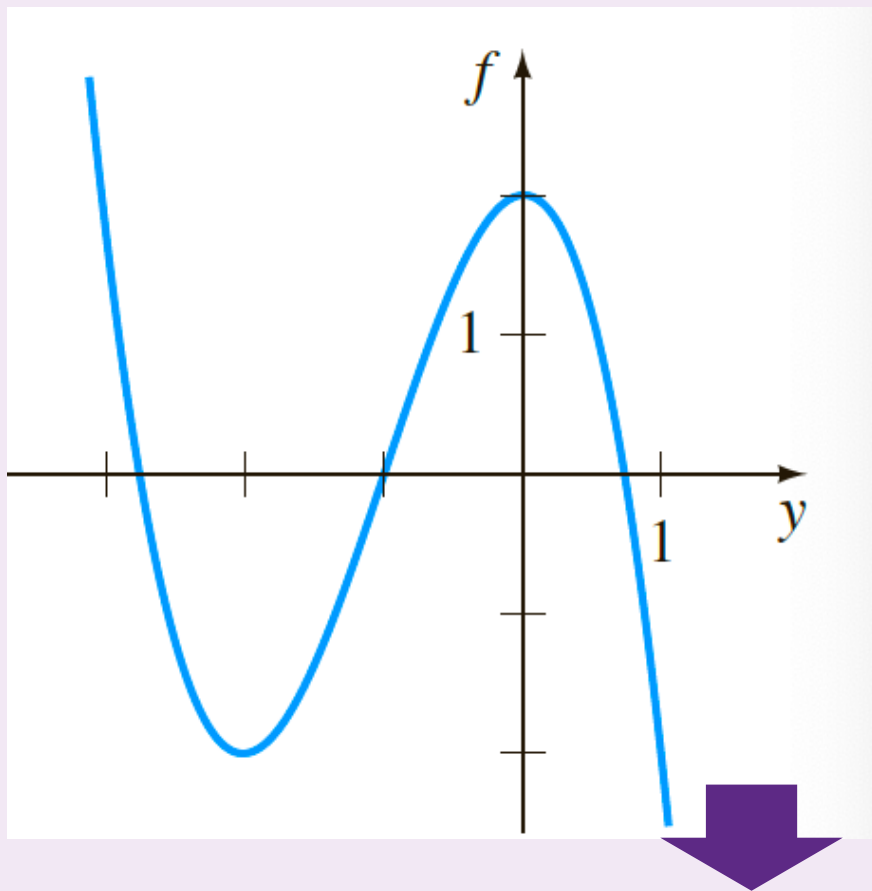
13.



$$\frac{dy}{dx} = f(y)$$



when $y \rightarrow -\infty \Rightarrow f(y) \rightarrow \infty$, means $y' \rightarrow \infty \uparrow$



when $y \rightarrow +\infty \Rightarrow f(y) \rightarrow -\infty$, means $y' \rightarrow -\infty \downarrow$

Summary

- Identify the sign of y' (you may draw a table or use some other techniques)
- When $y' > 0$, draw rising arrow ↗
- When $y' < 0$, draw falling arrow ↘
- When $y' = 0$, draw horizontal arrow → (it's a local max, or local min or reflection point)
- when $y' \rightarrow \infty$ ↑ when $y' \rightarrow -\infty$ ↓
- To guess the solution of IVP, first mark the initial point; by following the direction of segments, you can find the nearest approximate solution curve